Sampling from an infinite population can be handled by considering the distribution of the population. A random sample from an infinite population is considered as a random sample from a probability distribution. This idea will be used when we study testing of significance in the second year.

2.5 Census Method

The census method is also called complete enumeration method. In this method, information is collected from each and every individual in the statistical population.

Census of India is one of the best example. It is carried out once in every ten years. An enquiry is carried out, covering each and every house in India. It focuses on demographic details. They are collected and published by the Register general of India.

Appropriateness of this method:

The complete enumeration method is preferable provided the population is small and not scattered. Otherwise, it will have the following disadvantages.

Disadvantages:

- It is more time consuming, expensive and requires more skilled and trained investigators.
- More errors creep in due to the volume of work.
- Complete enumeration cannot be used if the units in the population destructive in nature. For example, blood testing, testing whether the rice is cooked or not in a kitchen,
- Testing the life times of bulbs etc.,
- When area of the survey is very large and there is less knowledge about the population, this method is not practicable. For example the tiger population in India, number trees in a forest cannot be enumerated using census method.

2.6 Sampling method:

In view of all these difficulties one has to resort to sampling methods for collecting thedata.

Sample is small proportion of the population taken from the population to study the characteristics of the population. By observing the sample one can make inferences about the population from which it is taken.

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